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Top Skills

OpenCV

Yolo

UNET

Languages

English

Bengali

Certifications

Triplebyte Certified

RCR Basic Course

Biomedical Responsible Conduct of Research Course.

Introduction to Large Language Models

Publications

COVID-19: Update, Forecast and Assistant - An Interactive Web Portal to Provide Real-Time Information and Forecast COVID-19 Cases in Bangladesh

Identifying the Stability of Couple Relationship Applying Different Machine Learning Techniques

Mining Significant Features of Diabetes Mellitus Applying Decision Trees: A Case Study In Bangladesh

Diseasome and comorbidities complexities of SARS-CoV-2 infection with common malignant diseases

Comorbidity Effects of Mitochondrial Dysfunction to the Progression of Neurological Disorders: Insights from a Systems Biomedicine Perspective

Koushik Howlader

NSF D4 NRT Fellow | Graduate Research Assistant | AI in Medicine
Ames, Iowa, United States

Summary

With over five years of experience in machine learning, I am seeking an intern/Co-op position that will enable me to continue my PhD for the next one or two years and research while contributing to innovative projects. I am particularly interested in roles that focus on research and development in areas such as machine learning, large language models (LLMs), deep learning, multi-modal systems, and testing of ML models. I do not require visa sponsorship. My goal is to leverage my expertise to advance cutting-edge solutions and drive impactful research.

Experience

Dependable Data Driven Discovery NRT (D4)

NSF D4 NRT Fellow

July 2025 - Present (1 year 2 months)

United States

CyMath at Iowa State University (ISU)

Mathematics Tutor

May 2025 - Present (1 year 4 months)

United States

Iowa State University

Graduate Research Assistant

August 2023 - Present (3 years 1 month)

Ames, Iowa, United States

I am working to develop a novel multi-modal contrastive pre-training framework that integrates X-ray images, ECG signals, and diagnostic reports using transformer-based models. We aim to enhance diagnostic accuracy and facilitate comprehensive patient assessments by aligning modality-specific features in a unified representation space.

Research Impact:

- I. Developed a robust framework that outperformed existing multimodal models in medical domain.
- ii. Demonstrated the framework's ability to generalize across different datasets and improve diagnostic accuracy with minimal labeled data.
- iii. Enhanced model interpretability, helping clinicians understand the reasoning behind the model's decisions through visual explanations.

North Dakota State University

2 years 8 months

Researcher-Facing Track CCAST Intern

May 2023 - August 2023 (4 months)

Fargo, North Dakota, United States

Responsibilities:

- i. Assisting researchers from various disciplines (e.g., biological sciences, engineering, computer science) effectively utilize High-Performance Computing (HPC) resources to streamline their workflows.
- ii. Facilitated the installation and optimization of scientific software on HPC systems, ensuring compatibility and performance for research tasks.
- iii. Worked with researchers to troubleshoot issues related to their code and provided optimization techniques to improve computational efficiency.
- iv. Developed user instructions, tutorials, and training materials to educate researchers on the proper use of shared HPC resources.
- v. Worked closely with faculty, researchers, and students from different fields, providing support and solutions for complex computational tasks.

Graduate Research Assistant

August 2021 - August 2023 (2 years 1 month)

United States

Advisor: Dr. Lu Liu

In this research project, we perform transfer learning pre-training dataset and fine-tuning effect analysis in cancer histopathology imaging by evaluating three popular deep neural network algorithms on three target datasets under various fine-tuning configurations. Pre-training models with cancer histopathology image datasets appear to perform worse or not better than pre-training models with ImageNet or random initialization. Furthermore, this study demonstrates that the performance of pre-trained models improves with the increase of images used in fine-

tuning, which was previously overlooked. Read full paper: DOI: 10.1109/BIBM55620.2022.9995076

Research Impact:

Demonstrated that using pre-trained models from ImageNet often outperformed those pre-trained on domain-specific medical datasets, challenging the assumption that domain-specific pre-training always yields better performance.

Mentor - REU Program

June 2023 - July 2023 (2 months)

United States

I had the distinct privilege of working as a mentor for the NSF Research Experiences for Undergraduates (REU) program. I wish to express my gratitude to my esteemed advisor, Dr. Lu, as well as to Dr. Simone Ludwig, Chair of the NDSU Department of Computer Science, for bestowing upon me this exceptional opportunity.

During the course of our investigation, our focus encompassed two discrete cell culture image datasets, denoted as MDA231(human breast cancer) and PC3(Bone Metastasis of a grade IV Prostatic Adenocarcinoma), both comprising sequential imagery spanning a triad of days. The image acquisition was executed employing a phase contrast microscope equipped with a 10X objective lens. Data pre-processing ensued, wherein patches were extracted from the original images employing a diverse array of methodologies including conventional cropping techniques and then segmentation techniques were implemented to segregate cellular entities from the background within the original images, delineating the background into black and white constituents. The overarching objective of our study revolved around the temporal characterization of cell cultures, facilitated through the utilization of deep learning methodologies such as transfer learning, few-shot learning, and conventional convolutional neural networks (CNNs).

Member - Graduate Council

July 2022 - June 2023 (1 year)

United States

President - Graduate Student Council

July 2022 - June 2023 (1 year)

United States

Instructor

January 2023 - May 2023 (5 months)

Fargo, North Dakota, United States

Teaching Assistant

January 2021 - May 2023 (2 years 5 months)

Tutor

August 2022 - December 2022 (5 months)

United States

SCRIBE - Graduate Student Council

June 2021 - May 2022 (1 year)

United States

Secretary General - Bangladeshi Student Organization (BSO)

June 2021 - May 2022 (1 year)

United States

MIET 2022

9 months

Technical Secretary

January 2022 - September 2022 (9 months)

Bangladesh

<https://confmiet.org/technical.php>

Session Moderator Chair

January 2022 - September 2022 (9 months)

Bangladesh

Reviewer

January 2022 - September 2022 (9 months)

Noakhali Science and Technology University

5 years 2 months

Assistant Professor (On Leave)

December 2017 - January 2021 (3 years 2 months)

Noakhali, Bangladesh

Lecturer

December 2015 - December 2017 (2 years 1 month)

Noakhali, Bangladesh

Here I have worked at the Department of Computer Science and
Telecommunication Engineering

IBCS-Primax Software Bangladesh Ltd.

Junior Programmer

May 2015 - November 2015 (7 months)

10/A, Dhanmondi, Dhaka

Developing

Infosys

Internship

January 2015 - April 2015 (4 months)

Mysore, Karnataka, India

Education

Iowa State University

Doctor of Philosophy - PhD, Computer Science · (August 2023 - December 2026)

North Dakota State University

Master's degree, Computer Science · (January 2021 - August 2023)

Jahangirnagar University

Master's Degree, Information Technology · (2014 - 2015)

Jahangirnagar University

Bachelor's Degree, Information Technology · (2010 - 2013)

Notre Dame College

Higher Secondary School, Science · (2006 - 2008)